Lecture 9

Advanced Quality Control

Types of Quality Control

In decreasing order of usage:

1. Trim reads by quality
2. Trim adapters from reads
3. Merge overlapping reads
4. Error correct reads
5. Remove duplicated data
6. Reduce excessive read coverage

Lower ranked tasks are used less frequently.

Sequence duplication

A controversial subject. Sequencing used to be an immature technology with substantial duplication errors. But the old rules don't apply anymore.

1. **Natural duplicates**: meaningful information
2. **Artificial duplicates**: misleading information

Natural duplication is a function of coverage. The more data you have, the more duplicates by chance.

Artificial duplicates (for example uneven PCR amplification) can produce strong, misleading signals.

What do I do with duplicates?

Some methods (SNP calling) are sensitive to the number of reads that indicate a signal. Better practices prescribe duplicate removal.

For other assays that deal with uneven read coverages: RNA-Seq, ChIP-Seq the consensus is to **not** remove duplicates.

High coverage methods will produce many natural duplicates that you need to know about for correct interpretation of the data.

How do I remove/mark duplicates?

1. **Sequence identity-based**: Find reads are identical and keep just one of them. Problem: rewards sequencing errors. Identical reads containing a sequencing mistake are kept.
2. **K-mer based** (sequence error correction): Find reads that contain sub-patterns that are seen only once in the data.
3. **Alignment based** (mark duplicates): Find reads that align over the same interval and collapse those into one. Can be used if there is an alignment file.

What is read merging?

During for paired end sequencing, if the DNA fragment is shorter than the sum of the read lengths then the reads may overlap:

```
------read1------>
========  DNA  =======
      <----read2------
```

The two opposing reads that have an overlapping region:

```
 ------read1------>
      ||||||||||||||
      <----read2------
```

Merge by detecting overlap

If we can detect the overlap:

```
------read1------>
      ||||||||||||
      <----read2------
```

We can combine two reads to produce a single, longer and more accurate read:

```
---------------------->
```

This is called "merging." Better data, fewer reads. A win-win.

Why would we merge reads?

Merge reads when the accuracy is important.

Merge reads when the when the fragment sizes are short!

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1. A recommended practice for 16S sequencing.
2. Merge if the majority of reads overlap.

Typically we want fragment sizes to be as long as possible! In those cases there is no overlap, hence no merging is possible.

How do I merge reads?

One of the most popular choices FLASH (Fast Length Adjustment of SHort reads):

```
source activate bioinfo
conda install flash -y
```

Get the 1% error test data from the tool website:

```
# Get an example dataset.
wget https://ccb.jhu.edu/software/FLASH/error1.tar.gz

# Unpack the data
tar zxvf error1.tar.gz
```

How does merging work?

Like all tools, it takes lots of parameters. By default run it as:

```
flash frag_1.fastq  frag_2.fastq
```

`flash` creates some ancillary files, while the report is:

```
[FLASH]    Total pairs:      1000000
[FLASH]    Combined pairs:   701191
[FLASH]    Uncombined pairs: 298809
[FLASH]    Percent combined: 70.12%
```

1% error rate has a 70% merge success!

How to make sense of many reports?

MultiQC to the rescue.

It is a tool that operates on the output of other tools.

Generates outputs that summarize multiple individual results.

Install with:

```
source activate bioinfo
conda install multiqc -y
```

Surprisingly many requirements.

How do I use MultiQC?

Get and unpack the data:

```
wget http://data.biostarhandbook.com/data/sequencing-platf
tar zxvf sequencing-platform-data.tar.gz
```

Run `fastqc` and make it keep the data directories

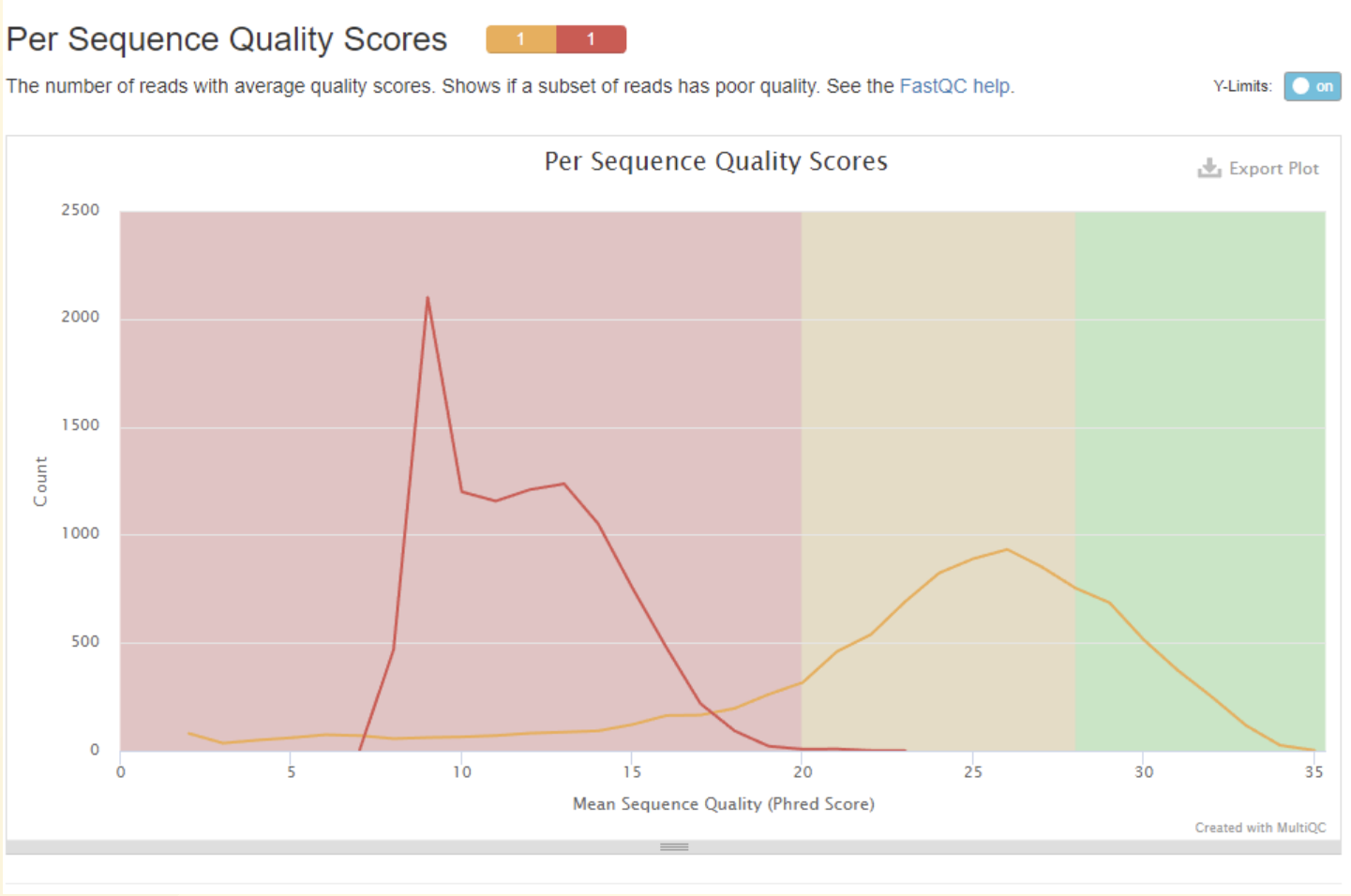
```
fastqc --extract illumina.fq iontorrent.fq
```

Run `multiqc` on the resulting directories:

```
multiqc illumina_fastqc iontorrent_fastqc
```

This, in turn, creates a file called `multiqc_report.html.`

MultiQC report



MultiQC supports many tools!

Not just FastQC.

You can learn a lot about what type of quality assessments may be done by looking at the tools that are supported.

A handy tool!

See the MultiQC website for example reports.

What is error correction?

Sequencing errors are (supposedly) random.

It is unlikely that they will occur in the same locations.

If the data has high coverage, we can recognize errors as the patterns that have the least support.

Error correction

Rationale (simplified scenario):

In this data, none of the reads support an `A`s at the same coordinate (vertical position).

```
TTATTTTTTTTTTTTTTTTTATTTTTTTT
TTTTTTTATTTTTTTTTTTTTTTTATTTT
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TTTTTTTTTTTTTTATTTTTTTTTTTTTT
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TTTTTTTTTTATTTTTTTTTTTTTTTTTT
TTTTTTTTTTTTTTTTTTTTTTTTTTTTT
TTTTTTTTTTTTTTTTTTTTTTTTATTTT
```

It is likely that all `A`s above were produced via sequencing instrument error.

How do I apply error correction?

The `bbmap` package includes the `tadpole.sh` error corrector:

```
tadpole.sh in=frag_1.fastq in2=frag_2.fastq
                mode=correct out=r1.fq out2=r2.fq
```

Produces:

```
Reads with errors detected: 1818127  (90.91%)
Reads fully corrected:      1555390  (85.55% of detected)
```

But the data we ran this one was claimed to have 2% error. This claims that `10%` of reads have errors...

Should I apply error correction?

We recommend error correction for assembly only.

But carefully evaluate before -> after.

We're not quite sure how well they work.

We are very skeptical!

They do "make" new data!

Brian Bushnell author of `bbtools` swears by them.

That's why we even have this section here 😀

Important to remember!

QC alters data! It is not without risk.

QC moves data toward a preconceived notion of what real data "ought" to look like.

Apply QC if your data needs it. Evaluate results before and after.

Adapter trimming, if done incorrectly, can destroy the meaning of your data!

Duplicate removal, if done incorrectly, can radically change your results!